

UBS Worst-of Memory Coupon Autocallable Note

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Executive Summary: We value the UBS worst-of memory coupon autocallable note by Monte Carlo simulation under the risk-neutral measure. The note is linked to Apple Inc., Amazon.com, Inc., and Alphabet Inc. Class A, with all triggers based on the worst-performing stock. The face amount is \$1,000, the quarterly contingent coupon is \$28.875, the call threshold is 100% of the initial price, and both the coupon barrier and downside threshold are 50% of the initial price. Using Bloomberg inputs for initial prices, USD OIS discount factors, dividend yields, 2-year ATM implied volatilities, 2-year 50% moneyness implied volatilities, and 251-point HRA correlations, the skew-aware Monte Carlo estimate is

$$V_0 = 959.30.$$

This is within about \$3.00 of UBS's reported estimated initial value of \$956.30. Compared with the flat-ATM volatility value of \$976.19, incorporating downside skew lowers the valuation by \$16.89, because higher downside volatility increases the probability of missed coupons and maturity principal loss.

1. Data

1.1 Contract terms. Let

$$S_1 = \text{AAPL}, \quad S_2 = \text{AMZN}, \quad S_3 = \text{GOOGL}.$$

The face amount and quarterly coupon are

$$N = 1000, \quad C = 28.875.$$

The initial prices are

$$S_{\text{AAPL},0} = 247.99, \quad S_{\text{AMZN},0} = 205.37, \quad S_{\text{GOOGL},0} = 301.00.$$

The contractual trigger levels are

$$B_{\text{call}} = 1.00, \quad B_{\text{coupon}} = 0.50, \quad B_{\text{down}} = 0.50.$$

The observation dates are quarterly from 22-Jun-2026 to 20-Mar-2028, and the payment dates are from 25-Jun-2026 to 23-Mar-2028.

1.2 Observation times and discount factors. Using valuation date 20-Mar-2026, the observation year fractions are

$$t = (0.2575, 0.5068, 0.7562, 1.0055, 1.2548, 1.5041, 1.7534, 2.0027),$$

and the interpolated USD OIS payment-date discount factors are

$$DF = (0.990222, 0.981223, 0.972467, 0.963927, 0.955597, 0.947347, 0.939443, 0.931605).$$

1.3 Bloomberg market inputs. The following inputs are mapped directly into the Monte Carlo model.

Input	Value	Role
$S_{AAPL,0}$	247.99	initial price / trigger denominator
$S_{AMZN,0}$	205.37	initial price / trigger denominator
$S_{GOOGL,0}$	301.00	initial price / trigger denominator
$\sigma_{AAPL,ATM}$	30.745%	normal-region volatility
$\sigma_{AMZN,ATM}$	38.031%	normal-region volatility
$\sigma_{GOOGL,ATM}$	37.397%	normal-region volatility
$\sigma_{50\%,AAPL}$	40.115%	downside-skew volatility
$\sigma_{50\%,AMZN}$	43.043%	downside-skew volatility
$\sigma_{50\%,GOOGL}$	43.012%	downside-skew volatility
q_{AAPL}	0.434%	risk-neutral drift adjustment
q_{AMZN}	0.000%	risk-neutral drift adjustment
q_{GOOGL}	0.288%	risk-neutral drift adjustment
$DF(0,t)$	USD OIS curve	cash-flow discounting

1.4 Correlation input. We use the 251-point Bloomberg HRA historical return correlation matrix as the baseline:

$$\rho_{251} = \begin{pmatrix} 1 & 0.546 & 0.495 \\ 0.546 & 1 & 0.518 \\ 0.495 & 0.518 & 1 \end{pmatrix},$$

with ordering ($AAPL, AMZN, GOOGL$). These are HRA R correlations, not beta estimates.

2. Valuation Setup

2.1 Product compression. Define the relative price

$$X_i(t) = \frac{S_i(t)}{S_i(0)}$$

and the worst-of ratio

$$W(t) = \min_{i=1,2,3} X_i(t).$$

The contract rules reduce to:

$$W(t_j) \geq 1 \Rightarrow \text{autocall},$$

$$W(t_j) \geq 0.5 \Rightarrow \text{coupon condition satisfied},$$

$$W(T) < 0.5 \Rightarrow \text{linear principal loss}.$$

2.2 Skew-aware risk-neutral dynamics. Under the risk-neutral measure,

$$\frac{dS_i(t)}{S_i(t)} = (r(t) - q_i) dt + \sigma_i(S_i, t) dW_i(t), \quad dW_i(t)dW_j(t) = \rho_{ij}dt.$$

To incorporate downside skew with minimal modification, we use a piecewise state-dependent volatility:

$$\sigma_i(S_i, t) = \begin{cases} \sigma_{50\%,i}, & S_i/S_{i,0} \leq 0.7, \\ \sigma_{ATM,i}, & S_i/S_{i,0} > 0.7. \end{cases}$$

Thus, the simulation uses ATM volatility in the normal region and 50% moneyness implied volatility after the stock enters a downside region. This is a simple local-volatility approximation using the newly collected

deep downside implied volatilities. In the simulation, the volatility used over $[t_k, t_{k+1}]$ is determined by the current simulated state $S_i(t_k)/S_{i,0}$, so the rule does not use future path information.

For each step,

$$S_i(t_{k+1}) = S_i(t_k) \exp \left[\left(r_k - q_i - \frac{1}{2} \sigma_i^2 \right) \Delta t_k + \sigma_i \sqrt{\Delta t_k} Z_{i,k} \right],$$

where

$$Z_k \sim N(0, \rho_{251}).$$

Correlated shocks are generated by

$$\rho = LL^\top, \quad Z = L\varepsilon, \quad \varepsilon \sim N(0, I_3).$$

2.3 Discounting. Cash flows are discounted on payment dates:

$$PV = \sum_{j=1}^8 DF(0, p_j) CF_j.$$

Forward rates between simulation dates are implied from discount factors:

$$r_k = -\frac{\ln(DF(0, t_{k+1})/DF(0, t_k))}{\Delta t_k}.$$

2.4 Memory coupon. Let M_j be the number of unpaid coupons before observation date t_j , with $M_0 = 0$. If

$$W(t_j) < 0.5,$$

then no coupon is paid and

$$M_j = M_{j-1} + 1.$$

If

$$W(t_j) \geq 0.5,$$

then the coupon payment is

$$C(M_{j-1} + 1),$$

and the memory balance resets to zero.

2.5 Autocall before maturity. For $j = 1, \dots, 7$, if

$$W(t_j) \geq 1,$$

then the note is autocalled and pays

$$CF_j = N + C(M_{j-1} + 1).$$

The path then terminates and contributes

$$PV_j = DF(0, p_j) [N + C(M_{j-1} + 1)].$$

2.6 Coupon without autocall. If

$$0.5 \leq W(t_j) < 1,$$

then the note pays

$$CF_j = C(M_{j-1} + 1),$$

but remains alive. If

$$W(t_j) < 0.5,$$

then

$$CF_j = 0,$$

and the missed coupon count increases.

2.7 Maturity payoff. If the note is never autocalled, then at maturity:

$$W(T) = \min_i \frac{S_i(T)}{S_i(0)}.$$

If

$$W(T) \geq 0.5,$$

then

$$CF_T = N + C(M_7 + 1).$$

If

$$W(T) < 0.5,$$

then the investor receives

$$CF_T = N W(T).$$

This captures the linear downside exposure to the worst-performing stock.

2.8 Monte Carlo estimator. For path n ,

$$PV^{(n)} = \sum_{j=1}^8 DF(0, p_j) CF_j^{(n)}.$$

The Monte Carlo estimator is

$$V_0 = \frac{1}{N_{\text{sim}}} \sum_{n=1}^{N_{\text{sim}}} PV^{(n)}.$$

The standard error is

$$SE = \frac{\text{Std}(PV^{(n)})}{\sqrt{N_{\text{sim}}}},$$

and the 95% confidence interval is

$$V_0 \pm 1.96SE.$$

3. Results Discussion

3.1 Skew-aware Monte Carlo result. Using 500,000 Monte Carlo paths, 251-point HRA correlation, ATM vol in the normal region, and 50% moneyness vol in the downside region, the value is

$$V_0 = 959.30.$$

The standard error is

$$SE = 0.3630,$$

and the 95% confidence interval is

$$[958.59, 960.01].$$

The result is very close to UBS's estimated initial value:

$$959.30 - 956.30 = 3.00.$$

Output	Value
Skew-aware MC price	959.30
Standard error	0.3630
95% confidence interval	[958.59, 960.01]
Autocall probability	50.97%
Maturity probability	49.03%
Principal loss probability at maturity	20.43%
Expected coupon paid, undiscounted	128.91
Expected principal paid, undiscounted	870.27
UBS estimated initial value	956.30
Difference vs UBS	3.00

3.2 Autocall profile. The skew-aware model gives the following autocall probabilities by observation date.

Observation date	Autocall probability
22-Jun-2026	24.19%
21-Sep-2026	9.70%
21-Dec-2026	5.78%
22-Mar-2027	4.05%
21-Jun-2027	2.99%
20-Sep-2027	2.35%
20-Dec-2027	1.90%

The first observation date has the largest autocall probability because the stocks begin at their initial levels. Later autocall probabilities decline because some paths have already been redeemed and because dispersion among the three underlyings accumulates.

3.3 Flat ATM versus skew-aware valuation. The skew-aware model materially lowers the value relative to the flat-ATM model.

Model	Price	Autocall Prob.	Principal Loss Prob.
Flat ATM volatility	976.19	50.42%	18.62%
Skew-aware volatility	959.30	50.97%	20.43%
Difference	-16.89	0.55%	1.81%

The price decreases because the 50% moneyness implied volatilities are higher than the ATM volatilities:

$$\sigma_{50\%,i} > \sigma_{ATM,i}.$$

Once a path enters the downside region, the higher volatility increases the probability of further adverse movement, missed coupons, and maturity principal loss. This is especially important for a worst-of note because only one stock needs to become the worst performer to reduce the payoff.

3.4 Interpretation. The previous flat-ATM model ignored downside skew:

$$\sigma_i(S_t/S_{i,0}) = \sigma_{ATM,i}.$$

That assumption undervalues tail risk and produces a higher price. The updated model uses

$$\sigma_i(S_i, t) = \begin{cases} \sigma_{50\%,i}, & S_i/S_{i,0} \leq 0.7, \\ \sigma_{ATM,i}, & S_i/S_{i,0} > 0.7, \end{cases}$$

which better reflects the fact that equity implied volatility is higher in downside states. The resulting value, \$959.30, is close to UBS's \$956.30. The remaining difference may come from UBS internal funding, dealer hedging costs, implied correlation assumptions, and more sophisticated volatility-surface modeling.

4. Conclusion

Using UBS product terms and Bloomberg market inputs, we construct a skew-aware Monte Carlo valuation for the worst-of memory coupon autocallable note. The payoff is driven by

$$W(t) = \min_i \frac{S_i(t)}{S_i(0)},$$

and the valuation formula is

$$V_0 = \mathbb{E}^Q \left[\sum_{j=1}^8 DF(0, p_j) CF_j \right].$$

After adding 2-year 50% moneyness implied volatilities, the model produces

$$V_0 = 959.30,$$

compared with the UBS estimated initial value of

$$956.30.$$

The updated result is much closer to UBS's value than the flat-ATM estimate of 976.19. This suggests that downside skew is a central pricing input for this structure, because coupon loss and principal loss are both driven by the 50% downside barrier region.

References: UBS AG, *Contingent Income Auto-Callable Securities with Memory Coupon linked to Apple Inc., Amazon.com, Inc., and Alphabet Inc. Class A*, pricing supplement (Form 424B2); Bloomberg Terminal inputs for AAPL, AMZN, and GOOGL spot prices, dividend yields, implied volatilities, HRA correlations, and USD OIS discount factors.